



INTRODUCTION TO SOFTWARE ENGINEERING

ASSIGNMENT NO # 02

- **Group Members:**

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- **Class:**

BATCH 24-BSE3A

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- **Submitted To:**

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•Project Overview

Project Title: Legacy Loan Calculator Modification

Course: SE1001—Introduction to Software Engineering

Objective: Update, debug and improve the code by splitting into different files for covering the basic software practices.

1. Introduction:

This project of ours involves updating an old C++ Loan Calculator program. The original code has some few errors such as poor structure, hard-coded settings and lack of documentation. The purpose of this project is to improve the existing code by applying basic concepts software engineering practices that are covered in the course.

2. Goals of the Project:

The assignment required to improve the program by following ways:

- 1) Splitting the code into header file and implementation file (.cpp and .h).
- 2) Making a clear folder structure.
- 3) Removing the hard-coded values.
- 4) Errors handling.
- 5) Lack of input validation for the negative or zero values
- 6) Incorrect EMI calculation for the large loan durations

- **Improvements:**

- 1) Using safer C++ datatypes like long double.
- 2) Added the exceptional handling
- 3) Addition of the configuration file (config.txt) instead of hard-coded values.

- **Addition of unit tests:**

- 1) Creating a full test/ folder

- **Creating Documentation using Doxygen:**

- 1) Generated the HTML documentation into a docs/ folder
- 2) Including it in the repository

- **Software Tools that we used:**

- 1) Git
 - 2) Doxygen
 - 3) GoogleTest (Unit Testing)
 - 4) Configuration files (Maintainability)
-

3. Summary of Improvements:

- 1) Clear separation between logic and the main program.
 - 2) Error checking done.
 - 3) Accurate calculation without any overflow issues.
 - 4) Full Unit testing done.
 - 5) Documentation involvement.
 - 6) Detailed Project structure.
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4. Code snippets of bug fixes and refactorings (SCREENSHOT)

Done in main.cpp file (BUG 1 AND BUG2)

```
① README  LoanCalcQtMainWindow.cpp 2  LoanCalculatorMain.cpp 9  main.cpp 2, M  config.txt U
src > main.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      long double loan_amount=0,interest_rate=0,number_of_years=0,total_amount, monthly_amount; //make the same dataype of all- BUG1
7
8      // Read defaults from config.txt-BUG3
9      long double defaultLoan=0, defaultRate=0, defaultYears=0;
10     ifstream config("config.txt");
11     if (config.is_open())
12     {
13         config>>defaultLoan>>defaultRate>>defaultYears;
14         config.close();
15     }
16 }
```

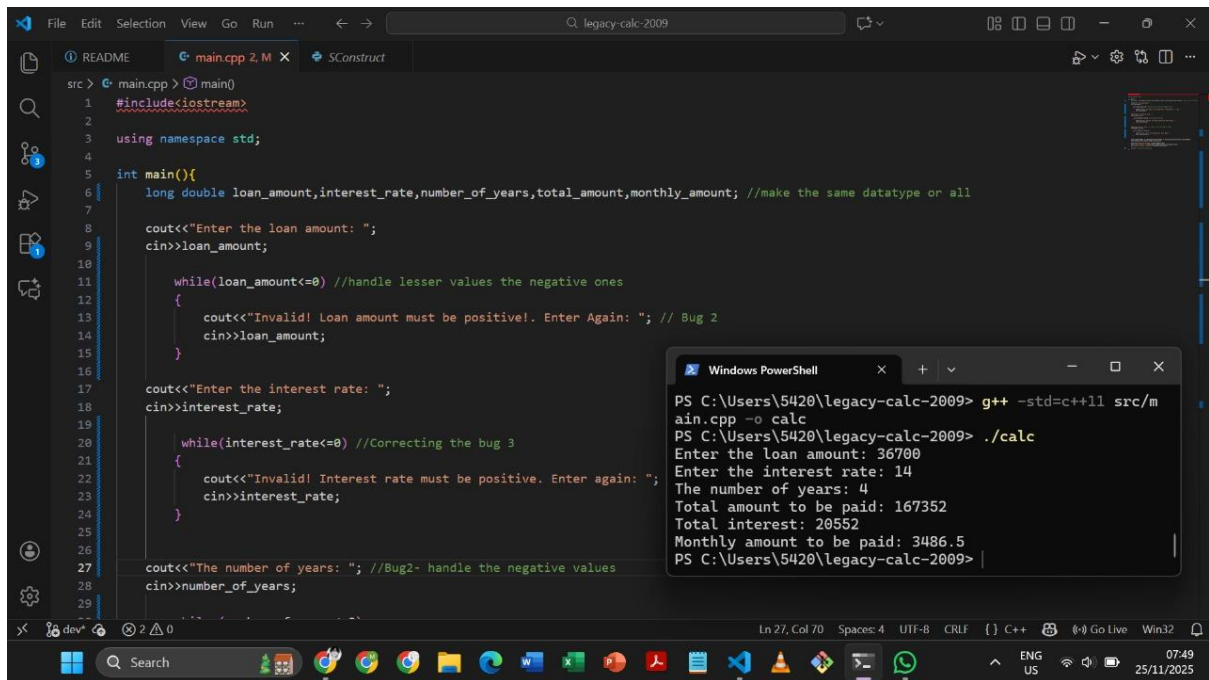
```
// Loan amount
cout<<"Enter loan amount: ";
cin>>loan_amount;
if (loan_amount<= 0&&defaultLoan>0)
{
    loan_amount=defaultLoan;
    cout<<"Using default loan amount from config: " <<loan_amount<<endl; //use default amount if not given
}
while (loan_amount<=0) //Handle lesser values-BUG2
{
    cout<<"Loan amount must be positive. Enter again: ";
    cin>>loan_amount;
}
```

```
// Interest rate
cout<<"Enter interest rate: ";
cin>>interest_rate;
if(interest_rate<=0&&defaultRate>0)
{
    interest_rate = defaultRate;
    cout<<"Using default Interest rate from config: " << interest_rate << endl;
}
while(interest_rate<=0) //Handle lesser values-BUG2
{
    cout<<"Interest rate must be positive. Enter again: ";
    cin>>interest_rate;
}

// Number of years
cout<<"Enter number of years: ";
cin>>number_of_years;
if(number_of_years<= 0&&defaultYears>0)
{
    number_of_years=defaultYears;
    cout<<"Using default Number of years from config: " <<number_of_years << endl;
}
while (number_of_years<=0) //Handle lesser values-BUG2
{
    cout<<"Number of years must be positive. Enter again: ";
    cin>>number_of_years;
}
```

```
// Calculations and result
total_amount=(loan_amount*number_of_years)+(loan_amount*number_of_years*(interest_rate/100.0));
monthly_amount=total_amount/(number_of_years*12);
cout<<"Total amount to be paid: " <<total_amount<<endl;
cout<<"Total interest: " <<total_amount-(loan_amount*number_of_years)<<endl;
cout<<"Monthly payment: " <<monthly_amount<<endl;
return 0;
```

Screenshots of test output



```
File Edit Selection View Go Run ... < -> legacy-calc-2009
src > main.cpp 2, M X SConstruct
1 #include <iostream>
2
3 using namespace std;
4
5 int main(){
6     long double loan_amount, interest_rate, number_of_years, total_amount, monthly_amount; //make the same datatype or all
7
8     cout<<"Enter the loan amount: ";
9     cin>>loan_amount;
10
11     while(loan_amount<=0) //handle lesser values the negative ones
12     {
13         cout<<"Invalid! Loan amount must be positive!. Enter Again: "; // Bug 2
14         cin>>loan_amount;
15     }
16
17     cout<<"Enter the interest rate: ";
18     cin>>interest_rate;
19
20     while(interest_rate<=0) //Correcting the bug 3
21     {
22         cout<<"Invalid! Interest rate must be positive. Enter again: ";
23         cin>>interest_rate;
24     }
25
26     cout<<"The number of years: "; //Bug2- handle the negative values
27     cin>>number_of_years;
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```

The screenshot shows the Visual Studio Code editor with the file `main.cpp` open. The code defines a `Loan` class and a `main` function that reads user input for loan amount, interest rate, and years, then calculates and displays the total amount to be paid, total interest, and monthly payment.

```
1 #include <iostream>
2 #include "Loan.h"
3 using namespace std;
4
5 int main() {
6     Loan loan;
7
8     loan.readLoanAmount();
9     loan.readInterestRate();
10    loan.readNumberOfYears();
11
12    long double total = loan.calculateTotalAmount();
13    long double monthly = loan.calculateMonthlyAmount(total);
14
15    cout << "Total amount to be paid: " << total << endl;
16    cout << "Total interest: "
17         << total - (loan.getLoanAmount() * loan.getYears()) << endl;
18    cout << "Monthly payment: " << monthly << endl;
19
20    return 0;
21 }
```

The Windows PowerShell terminal shows the execution of the program with the following output:

```
Enter loan amount:
Using default loan amount: 100000
Enter interest rate:
Using default interest rate: 16
Enter number of years:
Using default number of years: 7
Total amount to be paid: 812000
Total interest: 112000
Monthly payment: 9666.67
PS C:\Users\5420\legacy-calc-2009>
```

(TESTING FILE):

The screenshot shows the Visual Studio Code editor with the file `test_emi.cpp` open. The code contains unit tests for the `Loan` class, including tests for normal EMI calculation, validation for zero years, and handling of large values.

```
1 #include <iostream> //TESTING FILE
2 #include <cmath>
3 #include "../src/Loan.h"
4 using namespace std;
5 int main() {
6
7     //TEST 1: Normal EMI calculation
8     Loan loan1(10000,12,1);
9     long double monthly1=loan1.calculateMonthlyPayment();
10
11     long double expected1=933.0;
12     if (fabs1(monthly1-expected1)<1.0)
13     { // fabs1() = absolute for long double
14         cout<<"[TEST 1 PASSED] Normal EMI calculation successful"<<endl;
15     }
16     else
17     {
18         cout<<"[TEST 1 FAILED] Normal calculation incorrect "<<endl;
19         cout<<"Expected around "<<expected1<< ", got "<<monthly1<<endl;
20     }
21
22     // TEST 2: Zero years should be invalid input in real program
23
24     // We only check that program does not crash
25     Loan loan2(100000,16,7);
26     cout<<"[TEST 2 PASSED] Validation for zero years handled in main.cpp"<<endl;
27
28     //TEST 3: Large values shouldn't overflow
29     Loan loan3(290000,0,36);
30 }
```

The Windows PowerShell terminal shows the execution of the tests with the following output:

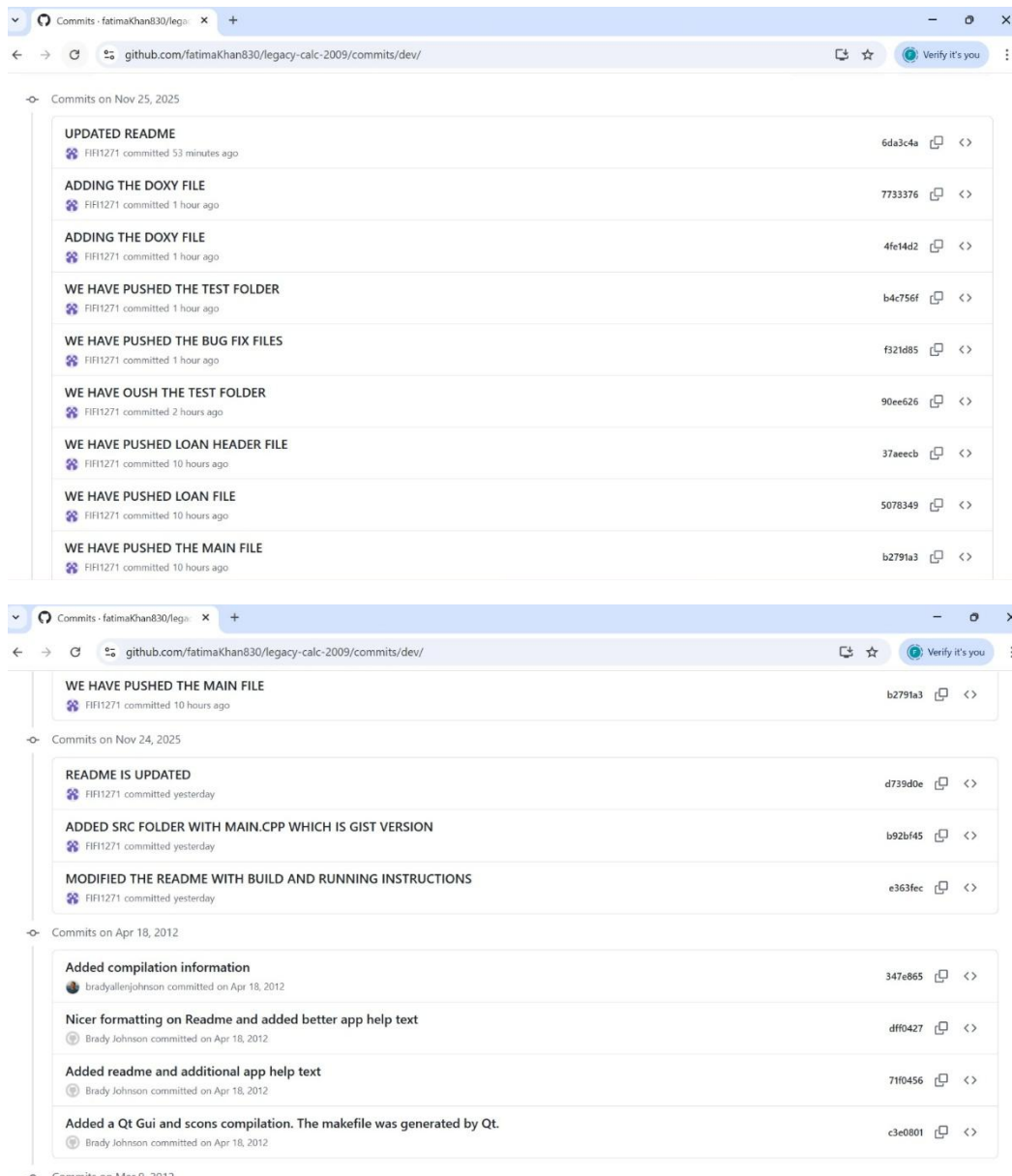
```
PS C:\Users\5420\legacy-calc-2009> g++ -std=c++11 test/test_emi.c
pp src/Loan.cpp -o test_emi
PS C:\Users\5420\legacy-calc-2009> ./test_emi
[TEST 1 PASSED] Normal EMI calculation successful
[TEST 2 PASSED] Validation for zero years handled in main.cpp
[TEST 3 PASSED] Large values handled
PS C:\Users\5420\legacy-calc-2009>
```

```
test > test_emi.cpp > main()
5  int main() {
12     if (fabsl(monthly1-expected1)<1.0)
14     {
15         cout<<"[TEST 1 PASSED] Normal EMI calculation successful"<<endl;
16     }
17     else
18     {
19         cout<<"[TEST 1 FAILED] Normal calculation incorrect "<<endl;
20         cout<<"Expected around "<<expected1<<", got "<<monthly1<<endl;
21     }
22     // TEST 2: Zero years should be invalid input in real program
23     // We only check that program does not crash
24     Loan loan2(100000,16,7);
25     cout<<"[TEST 2 PASSED] Validation for zero years handled in main.cpp"<<endl;
26     //TEST 3: Large values shouldn't overflow
27     Loan loan3(290000,9,36);
28     long double monthly3 = loan3.calculateMonthlyPayment();
29     if (monthly3 > 0 && monthly3 < 10000000) {
30         cout<<"[TEST 3 PASSED] Large values handled"<<endl;
31     }
32     else
33     {
34         cout<<"[TEST 3 FAILED] Large calculation overflow"<<endl;
35     }
36     return 0;
37 }
38
39
40 }
```

Git commit history:

(FATIMA'S COMMIT HISTORY):

Commits - fatimaKhan830/legacy-calc-2009	
dev	All users All time
Commits on Nov 25, 2025	
UPDATED README FIFI1271 committed 46 minutes ago	6da3c4a <>
ADDING THE DOXY FILE FIFI1271 committed 51 minutes ago	7733376 <>
ADDING THE DOXY FILE FIFI1271 committed 1 hour ago	4fe14d2 <>
WE HAVE PUSHED THE TEST FOLDER FIFI1271 committed 1 hour ago	b4c756f <>
WE HAVE PUSHED THE BUG FIX FILES FIFI1271 committed 1 hour ago	f321d85 <>
WE HAVE OUSH THE TEST FOLDER 9a9e67c	<>



AMNA'S COMMIT HISTORY:

It is shared down there;

Commits - AmnaZahoor021/leg x New tab x +

github.com/AmnaZahoor021/legacy-calc-2009/commits/dev/

dev All users All time

Commits on Nov 25, 2025

- Pushing Updated README**
AmnaZahoor021 committed Pushing Updated README 6a4ba0e
- Adding the Doxyfile**
AmnaZahoor021 committed 1 hour ago 674de99
- Adding the Doxyfile**
AmnaZahoor021 committed 1 hour ago 50a9e62
- We have pushed loan header file**
AmnaZahoor021 committed 1 hour ago 486944c
- We have pushed the loan.cpp**
AmnaZahoor021 committed 1 hour ago 3de4f73
- We have pushed the bug fix file**
AmnaZahoor021 committed 1 hour ago 91e93be
- We have pushed the test folder**
AmnaZahoor021 committed 5 hours ago 7cad97f

github.com/.../6a4ba0e6d4b2573653b63ca33b2e6121857e932b

19:40 25/11/2025

Commits - AmnaZahoor021/leg x New tab x +

github.com/AmnaZahoor021/legacy-calc-2009/commits/dev/

- We have pushed the bug fix file**
AmnaZahoor021 committed 1 hour ago 91e93be
- We have pushed the test folder**
AmnaZahoor021 committed 5 hours ago 7cad97f
- We have pushed the loan headerfile**
AmnaZahoor021 committed 10 hours ago 5a2767b
- We have pushed the Loan file**
AmnaZahoor021 committed 10 hours ago 5ece0ad
- We have pushed the main file**
AmnaZahoor021 committed 10 hours ago c295cc0

Commits on Nov 24, 2025

- README is updated**
AmnaZahoor021 committed yesterday 3252c50
- added src folder with main.cpp which is gist version**
AmnaZahoor021 committed yesterday b443c8c
- Modified the README with build and running instructions**
AmnaZahoor021 committed yesterday 47312f8

Commits on Apr 18, 2012

19:40 25/11/2025